Linear Time-Invariant Systems

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- LTI system is quite important
  - Most physical processes possess these two properties
  - It has been considerably analyzed.
- Superposition property
  - Decompose the very general input signals to linear combinations of basic signals or the delay of the basic signals
  - Output is the linear combination of the output of those basic signals
  - What basic signals? exponentials, sinusoids, unit impulse, etc

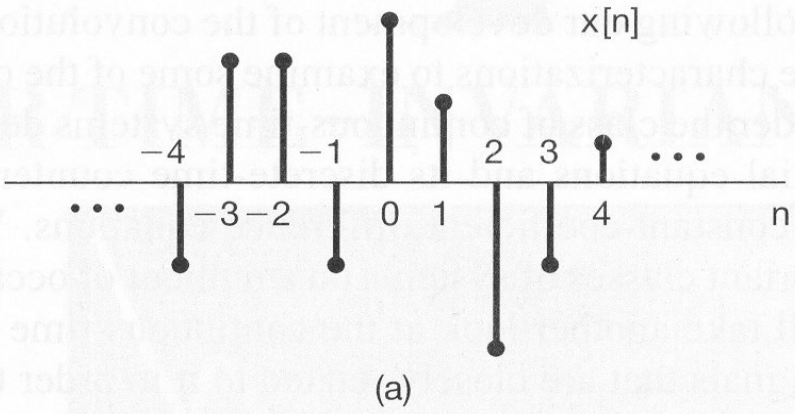
Example:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

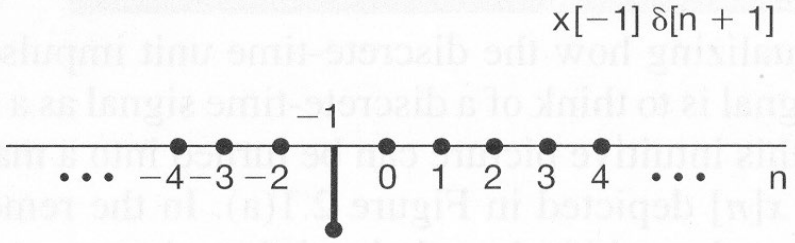
Discrete-Time Invariant Systems

- Representation of any $x[n]$ by impulses:

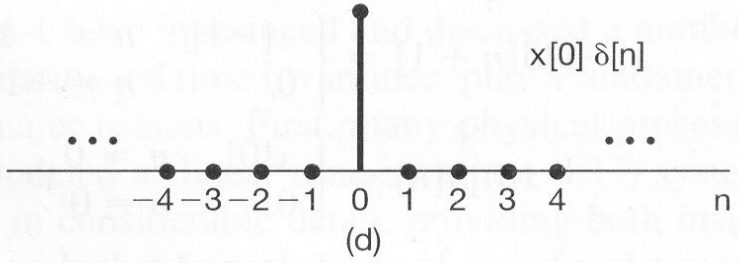
$$x[n] = \ldots + x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1] + \ldots$$



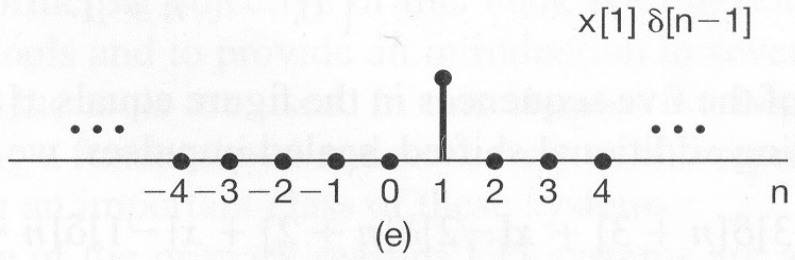
(a)



(c)



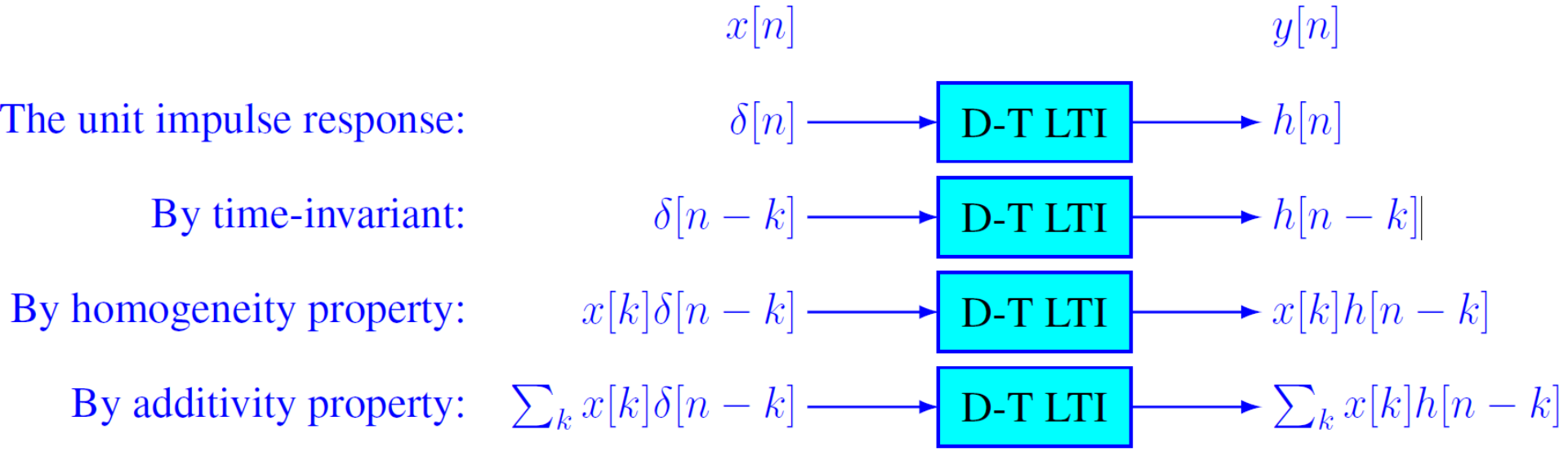
(d)



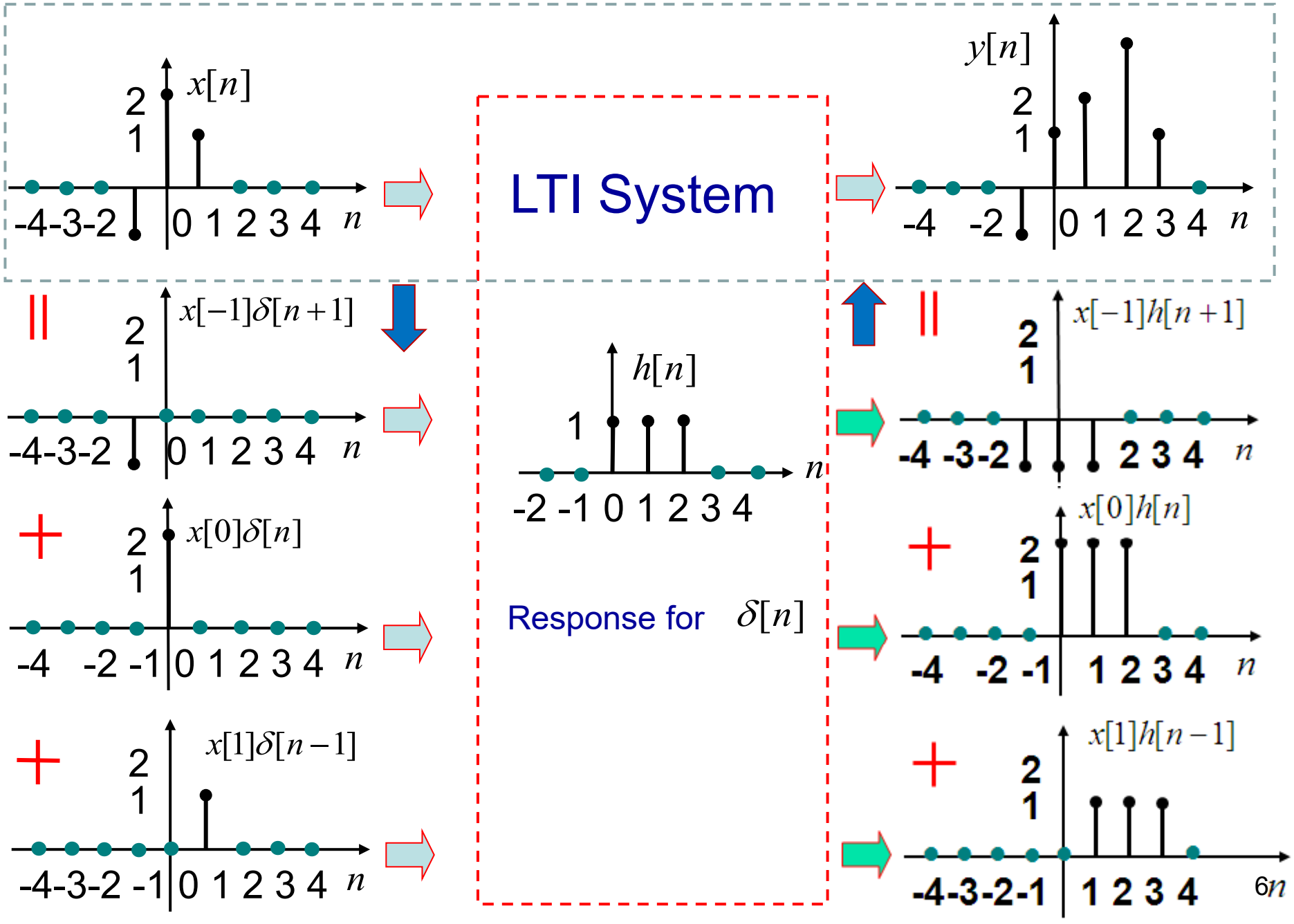
(e)

Check $x[n] = u[n]$

- Define unit impulse response as $h[n]$

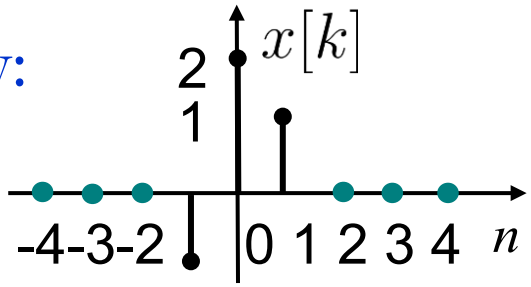


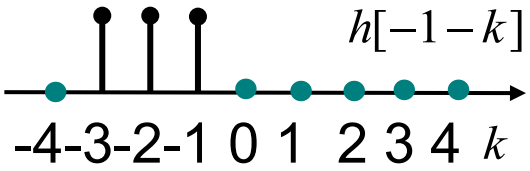
- Hence, $$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = x[n] * h[n]$$ Convolution sum
- LTI system is completely characterized by its unit impulse response

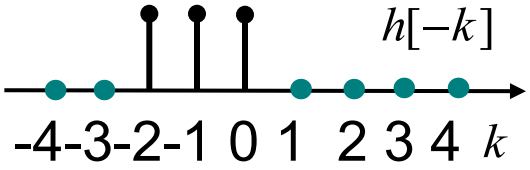

x[n]
2
1
-4-3-2
0 1 2 3 4
n
LTI System
y[n]
2
1
-4
-2
0 1 2 3 4
n
||
x[−1]δ[n+1]
x[−1]h[n+1]
h[n]
1
-2 -1 0 1 2 3 4
n
+
x[0]δ[n]
x[0]h[n]
Response for δ[n]
-4
-2 -1
+
x[1]δ[n−1]
x[1]h[n−1]
-4-3-2 -1
6n

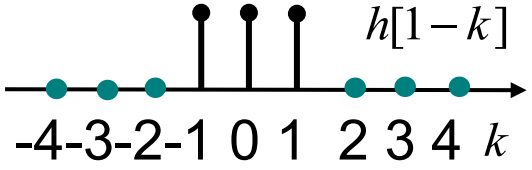
$$y[n_0] = \sum_{k=-\infty}^{\infty} x[k]h[n_0 - k]$$

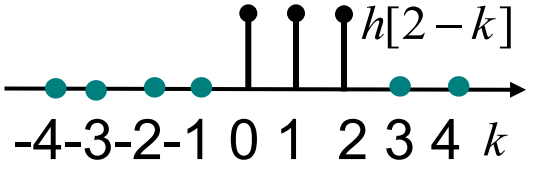
- More popular way:
  - For each particular n_0, view $y[n_0]$ as summation over functions of k
  - $h[n_0 - k]$ is the shift of $h[-k]$ to the right by n_0

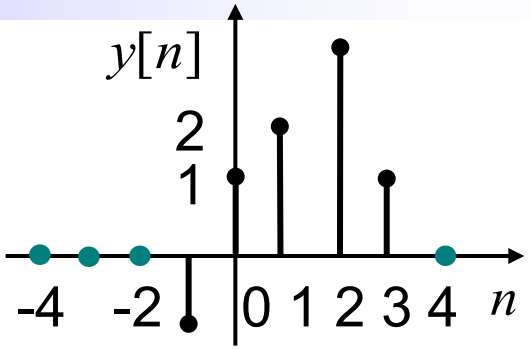












$$y[-1] = \sum_{k=-\infty}^{+\infty} x[k]h[-1-k] = -1$$

$$y[0] = \sum_{k=-\infty}^{+\infty} x[k]h[-k] = 1$$

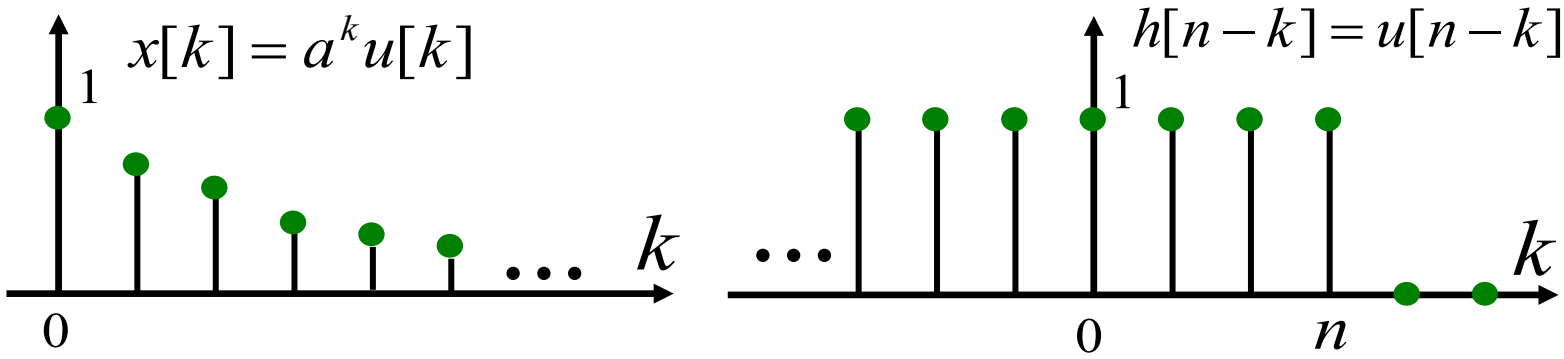
$$y[1] = \sum_{k=-\infty}^{+\infty} x[k]h[1-k] = 2$$

$$y[2] = \sum_{k=-\infty}^{+\infty} x[k]h[2-k] = 3$$

- Summary
  - Change the variable n to k and get $x[k]$ and $h[k]$
  - Perform time reversal of $h[k]$ and get $h[-k]$
  - Shift $h[-k]$ to the right by n_0 and get $h[-(k-n_0)] = h[n_0 - k]$
  - Multiple $x[k]$ and $h[n_0 - k]$, and sum over all values of k
  - You got $y[n_0]$

Actually, no need n_0 , you can directly use n

- Example:
  - $x[n] = \alpha^n u[n],\ 0 < \alpha < 1$ and $h[n] = u[n]$

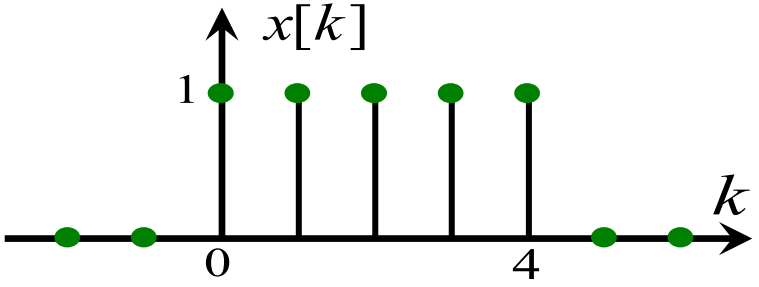


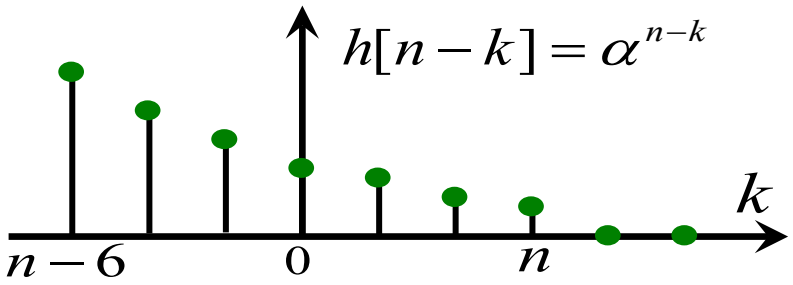
$$x[k]h[n-k] = \begin{cases} \alpha^k & 0 \le k \le n \\ 0 & \text{other } k \end{cases}$$

$$y[n] = \sum_{k=0}^{n} \alpha^k,\ n \ge 0 \quad \Longrightarrow \quad y[n] = \frac{1-\alpha^{n+1}}{1-\alpha} u[n]$$

– Example:

$$x[n] = \begin{cases} 1, & 0 \le n \le 4 \\ 0, & \text{otherwise} \end{cases} \qquad h[n] = \begin{cases} \alpha^n, & 0 \le n \le 6 \\ 0, & \text{otherwise} \end{cases}$$





$$n < 0, \quad y[n] = 0$$

$$0 \le n \le 4, \quad y[n] = \sum_{k=0}^{n} \alpha^{n-k},$$

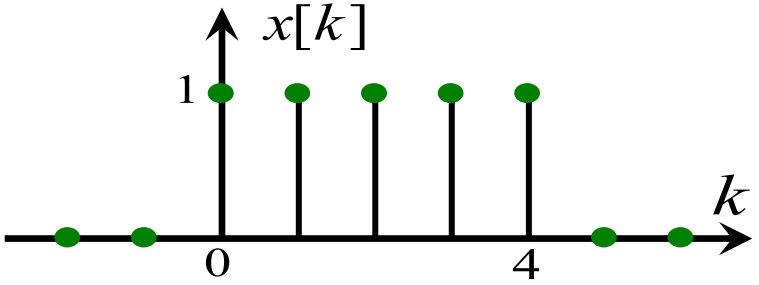
$$6 < n \le 10, \quad y[n] = \sum_{k=n-6}^{4} \alpha^{n-k}$$

$$4 < n \le 6, \quad y[n] = \sum_{k=0}^{4} \alpha^{n-k},$$

$$n > 10, \quad y[n] = 0$$

– Example:

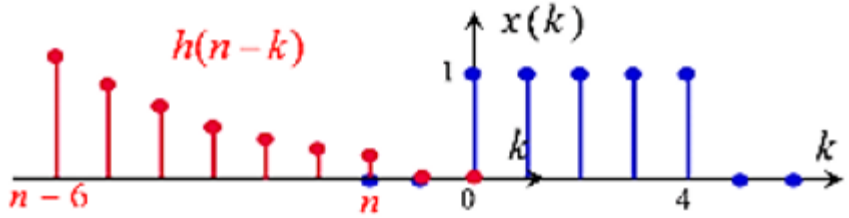
$$x[n] = \begin{cases} 1, & 0 \le n \le 4 \\ 0, & \text{otherwise} \end{cases}$$

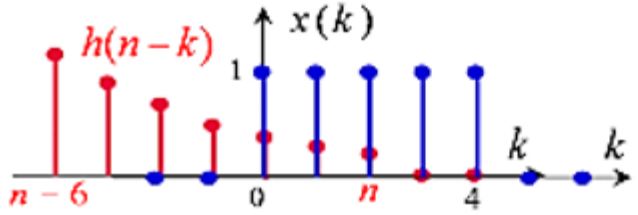


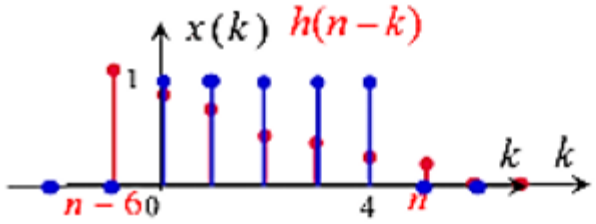
$$n < 0, \quad y[n] = 0$$

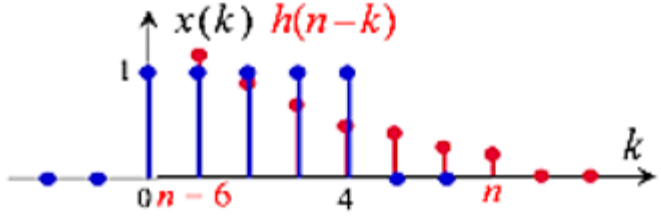
$$0 \le n \le 4, \quad y[n] = \sum_{k=0}^{n} \alpha^{n-k},$$

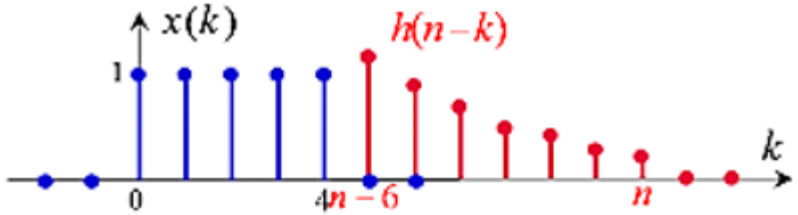
$$4 < n \le 6, \quad y[n] = \sum_{k=0}^{4} \alpha^{n-k},$$











$$\delta[n] \longrightarrow h[n]$$

$$x[k]\delta[n-k] \longrightarrow x[k]h[n-k]$$

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \longrightarrow y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

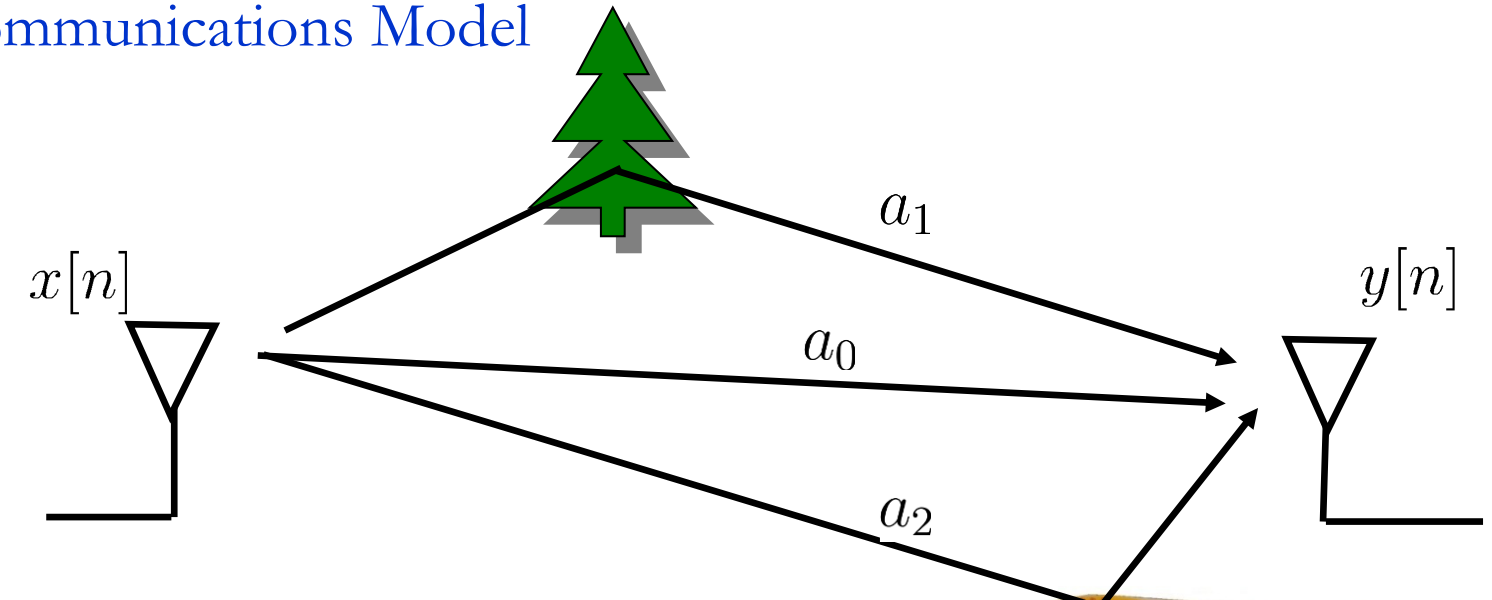
$$x[k]h[n_1 - k]$$

Output at a specific time n_1

$$y[n_1] = \sum_{k=-\infty}^{\infty} x[k]h[n_1 - k]$$

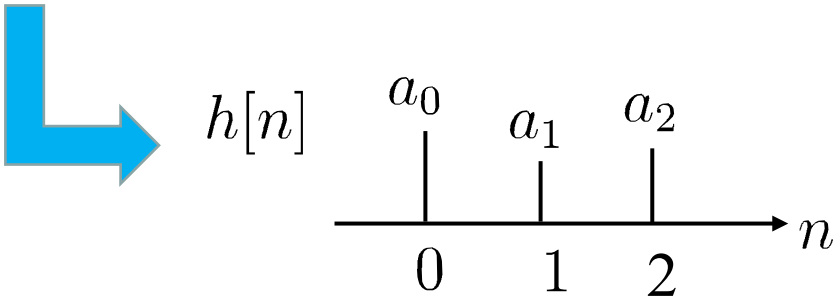
Physical Rationale of Convolution

- Communications Model



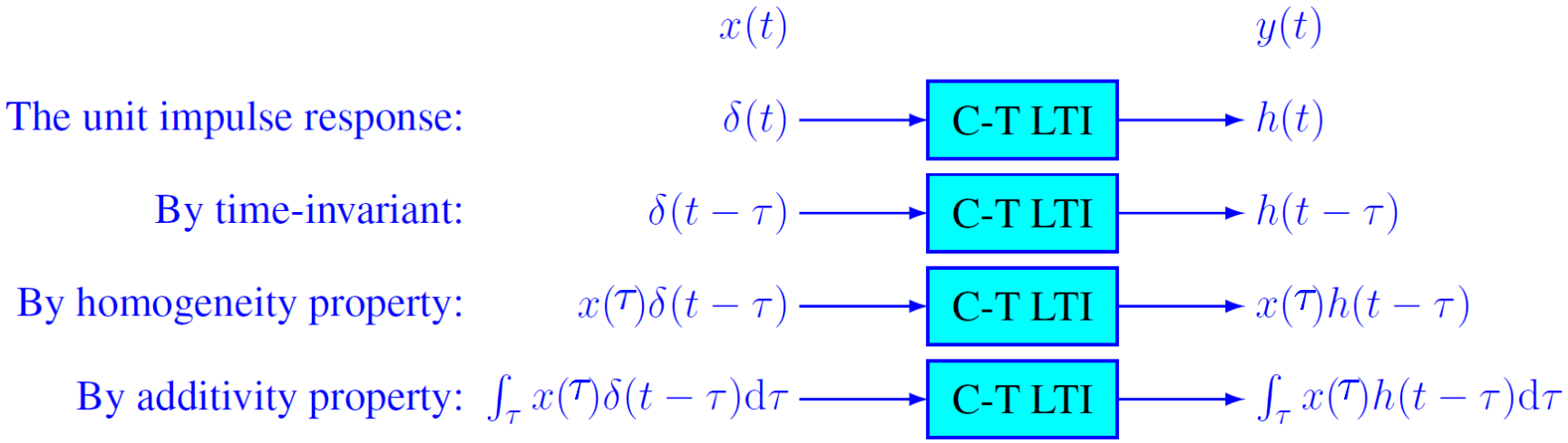
$$y[n] = a_0 x[n] + a_1 x[n-1] + a_2 x[n-2]$$





Continuous-Time LTI Systems

- A direct understanding (but not strict)



then $$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau = x(t) * h(t)$$

- For system with impulse response $\delta(t)$, the output is

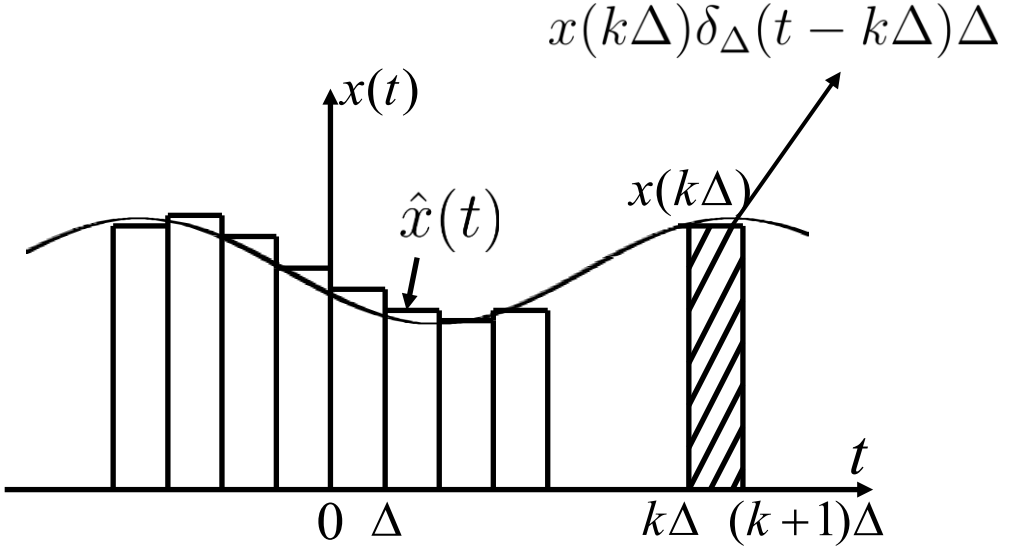
$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau$$

- Strict: staircase approximation of $x(t)$
  - Define

$$\delta_\Delta(t) = \begin{cases} \frac{1}{\Delta}, & 0 \le t \le \Delta \\ 0, & \text{otherwise} \end{cases}$$

The area is 1

$$\Delta \cdot \delta_\Delta(t) = \begin{cases} 1, & 0 \le t \le \Delta \\ 0, & \text{otherwise} \end{cases}$$



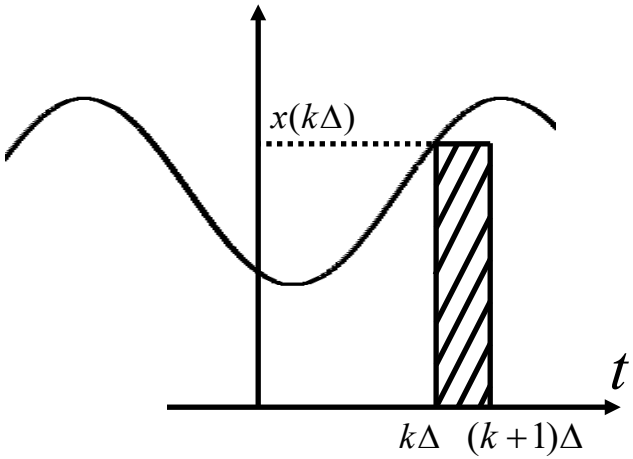
  - The k th square pulse is $x(k\Delta)\delta_\Delta(t-k\Delta)\Delta$
  - We can approximate $x(t)$ by

$$\hat{x}(t) = \sum_{k=-\infty}^{\infty} x(k\Delta)\delta_\Delta(t-k\Delta)\Delta$$

Continuous-Time LTI Systems

- As $\Delta \to 0$, $\hat{x}(t) \to x(t)$

$$x(t) = \lim_{\Delta \to 0} \sum_{k=-\infty}^{\infty} x(k\Delta)\delta_{\Delta}(t - k\Delta)\Delta$$



Consider $k\Delta$ as τ , and Δ as $d\tau$

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t - \tau)d\tau$$

On the other side, Remember $x(\tau)\delta(t-\tau) = x(t)\delta(t-\tau)$

$$\int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau = \int_{-\infty}^{\infty} x(t)\delta(t-\tau)d\tau = x(t)\int_{-\infty}^{\infty} \delta(t-\tau)d\tau = x(t)$$

Hence, we may still treat the continuous-time signals as the weighted “sum” of shifted unit impulse

- Define $h_\Delta(t)$ as the response to $\delta_\Delta(t)$
  - From time invariance, input $\delta_\Delta(t-k\Delta)$ yields $h_\Delta(t-k\Delta)$
  - For LTI system, the impulse representation of output is

$$\hat{y}(t) = \sum_{k=-\infty}^{\infty} x(k\Delta) h_\Delta(t-k\Delta)\Delta$$

  - Define $h(t) \overset{\Delta\to 0}{=} h_\Delta(t)$, i.e., the response to $\delta(t)$
  - As $\Delta \to 0$, then $\hat{x}(t) \to x(t)$, and $\hat{y}(t) \to y(t)$

$$y(t) = \lim_{\Delta\to 0} \hat{y}(t) = \lim_{\Delta\to 0} \sum_{\infty} x(k\Delta) h_\Delta(t-k\Delta)\Delta$$

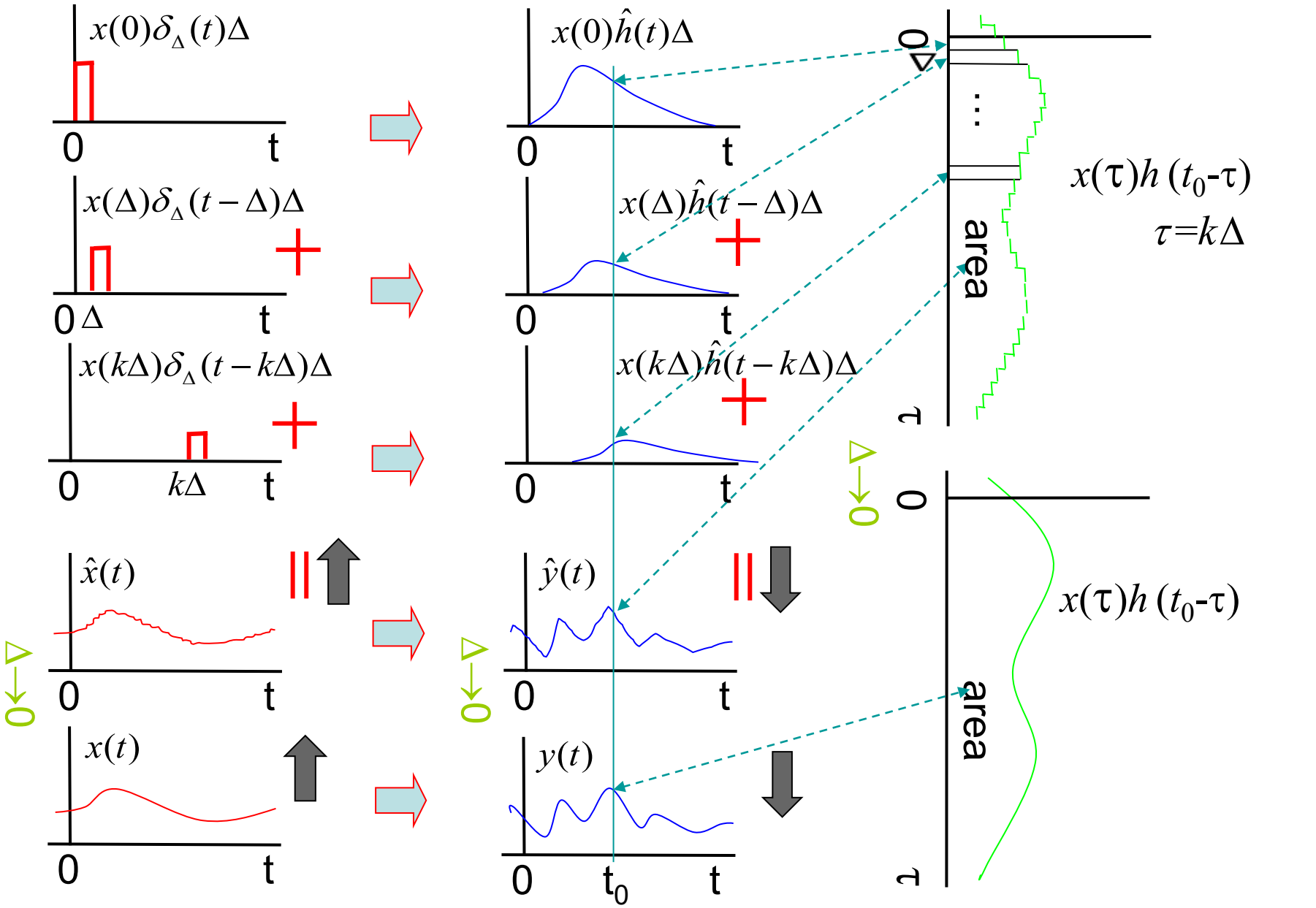
$$= \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = x(t) * h(t)$$

Convolution integral

Both the continuous and discrete-time convolutions are completely characterized by their impulse responses

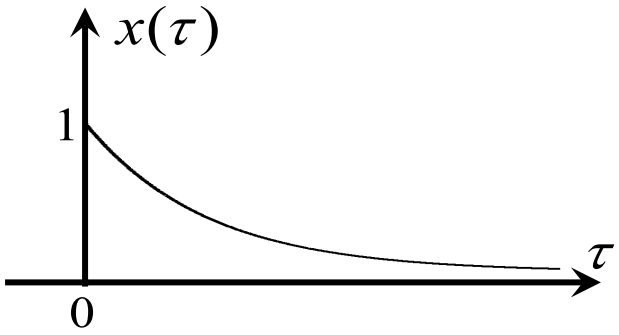
$$\hat{x}(t) = \lim_{\Delta \to 0} \sum_{k=-\infty}^{\infty} x(k\Delta)\delta_{\Delta}(t - k\Delta) \cdot \Delta$$

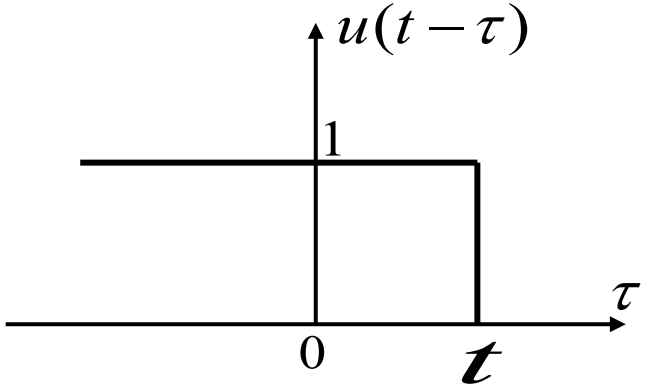
$$\hat{y}(t_0) = \sum_{k=-\infty}^{\infty} x(k\Delta)\hat{h}(t_0 - k\Delta) \cdot \Delta$$

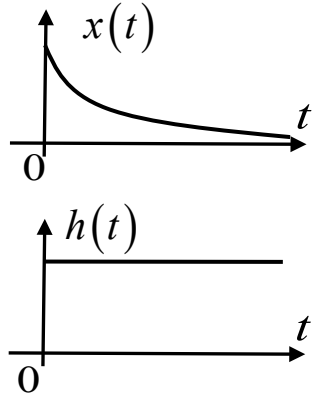


Computation of Convolutional Integral

- Similar as in discrete-time case, consider $x(\tau)$ and $h(t-\tau)$ as functions of τ.
  - $h(t_0-\tau)$ is the shift of $h(-\tau)$ to the right by t_0
  - Multiplication and integration over τ yields $y(t)$ (area)
  - Example $x(t)=e^{-at}u(t), \quad a>0 \qquad h(t)=u(t)$



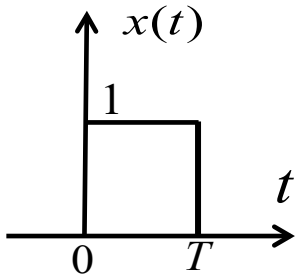


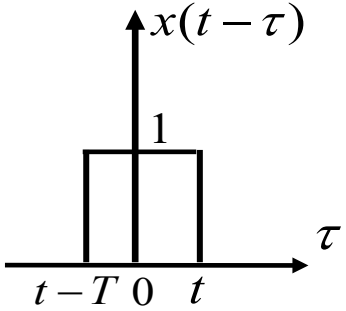


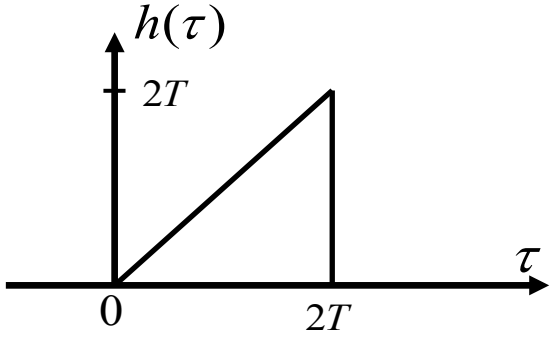
$$y(t)=\int_{-\infty}^{+\infty}x(\tau)h(t-\tau)d\tau=\frac{1}{a}(1-e^{-at})u(t)$$

– Example

$$x(t)=\begin{cases}1, & 0<t<T\\ 0, & \text{otherwise}\end{cases} \qquad h(t)=\begin{cases}t, & 0<t<2T\\ 0, & \text{otherwise}\end{cases}$$







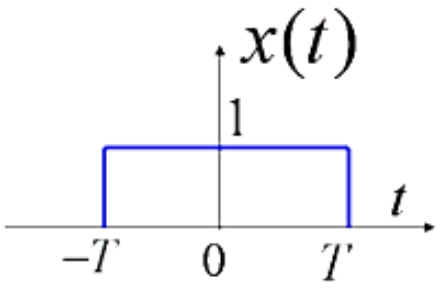
$$y(t)=x(t)*h(t)=\int_{-\infty}^{\infty}x(\tau)h(t-\tau)d\tau=\int_{-\infty}^{\infty}x(t-\tau)h(\tau)d\tau$$

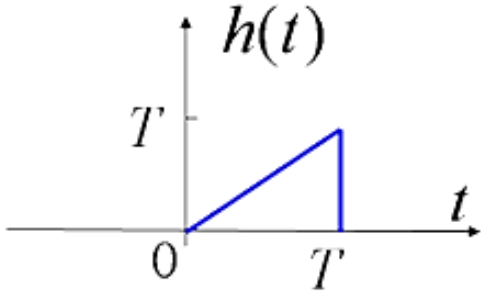
$y(t)=0 \qquad t<0 \qquad t>3T$

$$y(t)=\int_0^t \tau d\tau=\frac{1}{2}t^2 \qquad 0<t<T$$

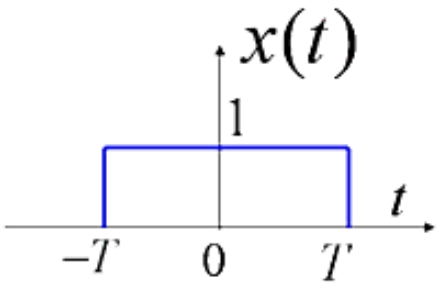
$$y(t)=\int_{t-T}^{t}\tau d\tau=Tt-\frac{1}{2}T^2 \qquad T<t<2T$$

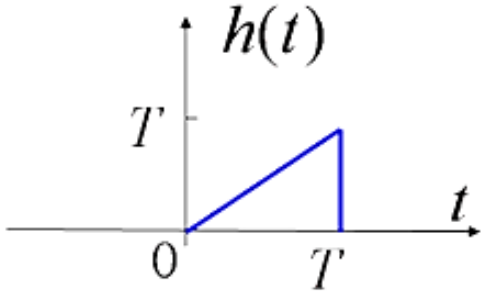
$$y(t)=\int_{t-T}^{2T}\tau d\tau=2T^2-\frac{1}{2}(t-T)^2 \qquad 2T<t<3T$$





Exercise

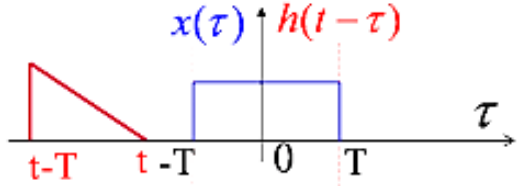


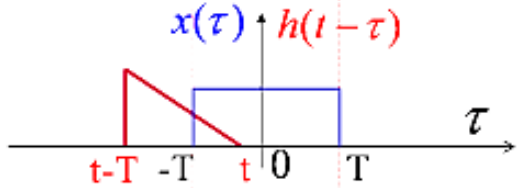


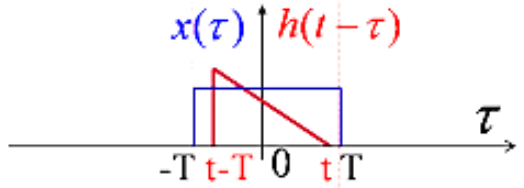
1. $t < -T, y(t) = 0$
2. $-T < t < 0, (t - T < -T)$

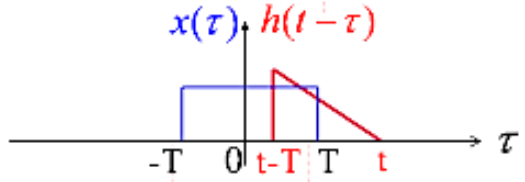
$$y(t) = \int_{-T}^{t} (t - \tau) d\tau = \frac{1}{2}t^2 + tT + \frac{1}{2}T^2$$

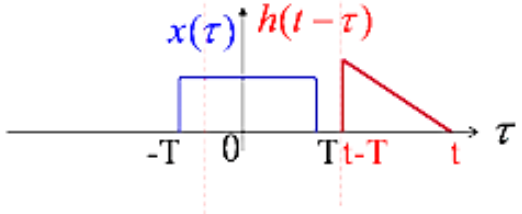
3. $0 < t < T, (-T < t - T < 0)$
   $y(t) = \int_{t-T}^{t} (t - \tau) d\tau = \frac{1}{2}T^2$
4. $T < t < 2T, (0 < t - T < T)$
   $y(t) = \int_{t-T}^{T} (t - \tau) d\tau = -\frac{1}{2}t^2 + tT$
5. $t > 2T, (t - T > T), y(t) = 0$











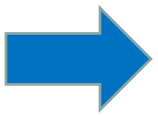
Property of LTI Systems

- An LTI system is completely defined by its impulse response. It's important to emphasize that this property holds in general only for LTI systems.

$$y[n] = x[n] + x[n-1]$$

$$y[n] = (x[n] + x[n-1])^2$$

$$y[n] = \max(x[n], x[n-1])$$



$$h[n] = \begin{cases} 1, & n = 0, 1 \\ 0, & \text{otherwise} \end{cases}$$

Property of LTI Systems

- The Commutative Property

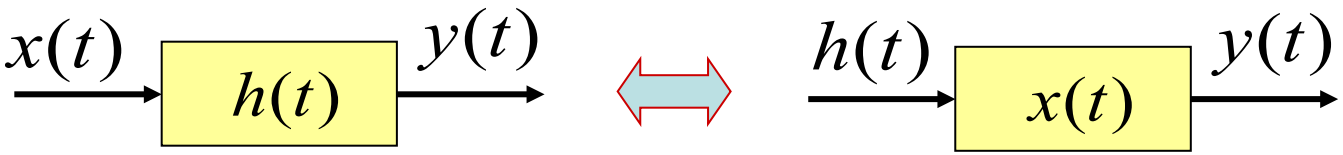
$$x[n] * h[n] = h[n] * x[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

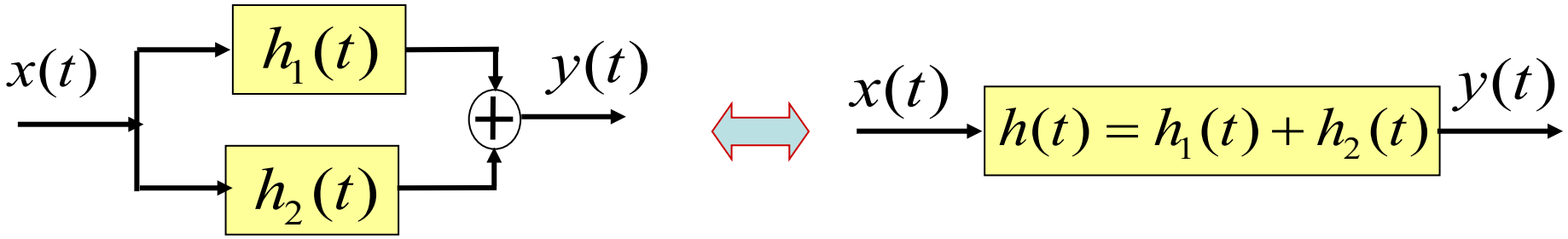
$$x(t) * h(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$$

- The Distributive Property

$$x(t) * [h_1(t) + h_2(t)] = x(t) * h_1(t) + x(t) * h_2(t)$$

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$





– Combination of commutative and distributive

$$(x_1[n] + x_2[n]) * h[n] = x_1[n] * h[n] + x_2[n] * h[n]$$

$$(x_1(t) + x_2(t)) * h(t) = x_1(t) * h(t) + x_2(t) * h(t)$$

– Example

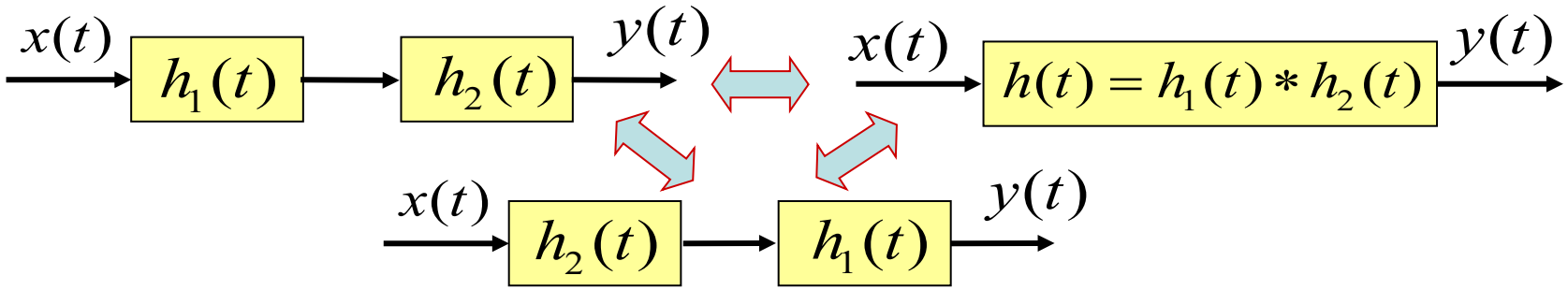
$$x[n] = (2)^{-n}\,u[n] + 2^n u[-n] \qquad h[n] = u[n]$$

$$y[n] = (4 - 2^{-n})u[n] + 2^{n+1}u[-n-1]$$

- The Associative Property

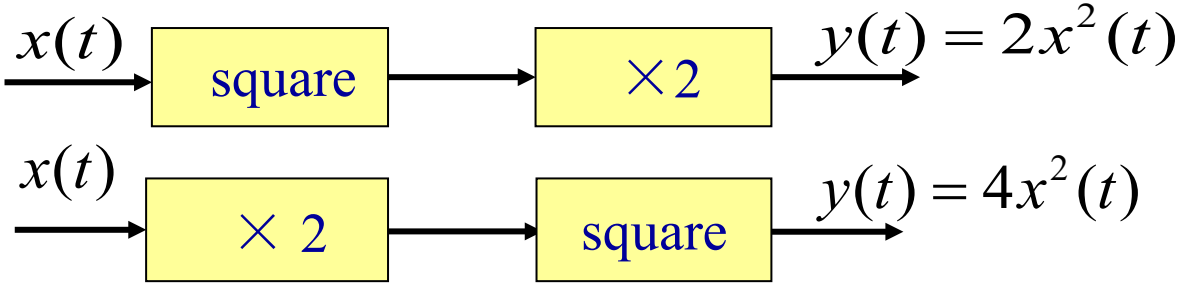
$$x[n] * (h_1[n] * h_2[n]) = (x_1[n] * h_1[n]) * h_2[n]$$

$$x(t) * (h_1(t) * h_2(t)) = (x_1(t) * h_1(t)) * h_2(t)$$



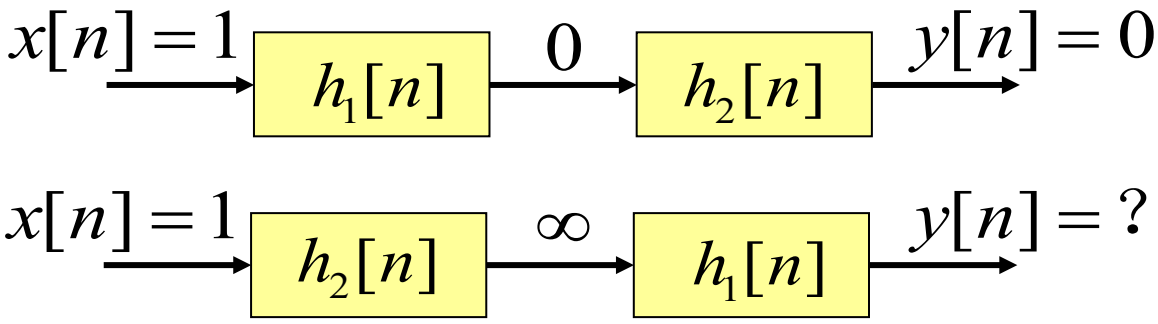
  - The impulse response of the cascade of two LTI systems is the convolution of their individual impulse responses.
  - The unit impulse response of a cascade of two LTI systems does not depend on the order in which they are cascaded.

- The associate property has particular requirements:
  - The system must be LTI systems



  - Any convolution operation included must converge

$$h_1[n] = \delta[n] - \delta[n-1], \quad h_2[n] = u[n] \qquad x[n] = 1$$



- Other properties of convolution
  - $x(t) * \delta(t) = x(t)$

    $x(t) * \delta(t - t_0) = x(t - t_0)$

    $x(t - t_1) * \delta(t - t_2) = x(t - t_1 - t_2)$

  - If $x(t) * h(t) = y(t)$, then

    $x(t - t_1) * h(t - t_2) = y(t - t_1 - t_2)$

  - $x(t) * u(t) = \int_{-\infty}^{t} x(\tau) d\tau$

Different from multiplication

- The properties of the LTI systems are reflected by the properties of its impulse response
- LTI with and without Memory
  - Discrete LTI is memoryless if $h[n] = 0$ for $n \neq 0$

$$h[n] = K\delta[n] \qquad y[n] = Kx[n]$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

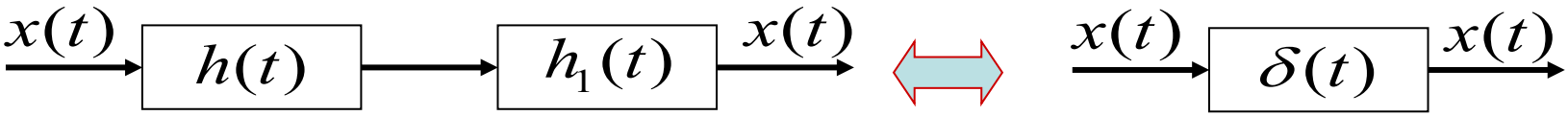
  - Continuous LTI is memoryless if $h(t) = 0$ for $t \neq 0$

$$h(t) = K\delta(t) \qquad y(t) = Kx(t)$$

  - If $K = 1$, then the system is identity system

Check $x(t) * \delta(t)$!!!

- Invertability



  - The invertible system must has the impulse response with

$$h(t) * h_1(t) = \delta(t) \qquad h[n] * h_1[n] = \delta[n]$$

  - Example: what is the inverse system for $y(t) = x(t - t_0)$? Check the impulse response first.
  - Example: Consider the system with $h[n] = u[n]$

$$y[n] = \sum_{k=-\infty}^{n} x[k] \qquad x[n] = y[n] - y[n-1]$$

$$\Rightarrow h_1[n] = \delta[n] - \delta[n-1]$$

$$u[n] * (\delta[n] - \delta[n-1]) = u[n] - u[n-1] = \delta[n]$$

- Causality
  - An LTI is causal if $h[n] = 0, n < 0$, or $h(t) = 0, t < 0$.
  - If the input to a causal system is 0 up to some point in time, then the output must also be 0 up to that time→*Condition of initial rest.*
  - $y[n] = 2x[n] + 3$, casual but does not satisfy the initial rest
  - The accumulator? Causal
  - The inverse of accumulator? Causal
  - The shifter? Causal?
  - The inverse of shifter? Non-Causal?
  - Alternative expression: $y[n] = \sum_{k=0}^{+\infty} h[k]x[n-k]$

- Stability: when bounded input lead to bounded output

$$|y[n]| = \left| \sum_{k=-\infty}^{\infty} h[k]x[n-k] \right| \leq \sum_{k=-\infty}^{\infty} |h[k]||x[n-k]|$$

  - For $|x[n]| \leq B \Rightarrow |y[n]| \leq B \sum_{k=-\infty}^{\infty} |h[k]|$
  - The impulse response should be absolutely summable:

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

Necessary condition too

  - Absolutely integrable: The continuous-time LTI is stable iff

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$$

Identity system?
Accumulator?

- For any stable $h[n]$, counter-assume $h[n]$ not absolutely summarable.
  - Define $$x[n] = \begin{cases} 0 & h[-n] = 0 \\ \dfrac{h^*[-n]}{|h[-n]|} & h[-n] \neq 0 \end{cases}$$

  We know $|x[n]| \leq 1$ is bounded
  - Check $$y[0] = \sum_{k=-\infty}^{\infty} h[k]x[0-k] =$$
  $$= \sum_{\substack{k=-\infty \\ h[k]\neq 0}}^{\infty} h[k]\frac{h^*[k]}{|h[k]|} = \sum_{n=-\infty}^{\infty} |h[n]| = \infty$$
  - Hence, the output is unbounded, and the counter-assumption is wrong.

Unit Step Response of LTI

- Unit step response can also describe LTI system
  - Input $x[n] = u[n],\ x(t) = u(t)$
  - Response

$$s[n] = u[n] * h[n] = \sum_{k=-\infty}^{n} h[k]$$

$$s(t) = u(t) * h(t) = \int_{k=-\infty}^{t} h(\tau)d\tau$$

  - Relationship with impulse response

$$h[n] = s[n] - s[n-1] \qquad h(t) = \frac{ds(t)}{dt} = s'(t)$$

LTI System Described by Differential Equations

- A wide variety of systems and physical phenomena can be described by the linear constant-coefficient differential equations (LCCDE)

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}$$

a_k, b_k constants

  - Implicit specification of the systems rather than explicit expression, just the relationship or a constraint

What if $N = 0$?

  - Particular solution plus a homogeneous solution

where

$$y(t) = y_p(t) + y_h(t)$$

Natural response

$$\sum_{k=0}^{N} a_k \frac{d^k y_h(t)}{dt^k} = 0$$

Still not determined

- To gain explicit expression of the output, solve the equation with one or more auxiliary conditions (many choices).
- For causal LTI in this book, initial rest condition holds

$$y(t_0) = \frac{dy(t_0)}{dt} = \ldots = \frac{d^{N-1}y(t_0)}{dt^{M-1}} = 0$$

- Example: $\frac{dy(t)}{dt} + 2y(t) = x(t)$

  input signal $x(t) = 4e^{3t}u(t)$

  homogeneous equation $\frac{dy_h(t)}{dt} + 2y_h(t) = 0$

$$\frac{dy(t)}{dt} + 2y(t) = x(t)$$

– A common method to find $y_p(t)$ for exponential input is – *forced response*.

– We hypothesize $y_p(t) = Ye^{3t}, \quad t > 0$, then

$$3Ye^{3t} + 2Ye^{3t} = 4e^{3t} \quad \Rightarrow Y = \frac{4}{5} \qquad \Rightarrow y_p(t) = \frac{4}{5}e^{3t}u(t)$$

– To determine $y_h(t)$, we hypothesize for $t > 0$

$$y_h(t) = Ae^{st}$$

then

$$Ase^{st} + 2Ae^{st} = Ae^{st}(s+2) = 0 \quad \Rightarrow s = -2$$

The solution for $t > 0$ is

$$y(t) = Ae^{-2t} + \frac{4}{5}e^{3t}$$

The solution is still not explicit!!!

$$y(t) = Ae^{-2t} + \frac{4}{5}e^{3t}$$

– For causal system, the auxiliary conditions can be taken from the initial rest.

$$x(t) = 0, \quad t < t_0 \quad \Rightarrow \quad y(t) = 0, \quad t < t_0$$

Setting $y(0) = 0$ gives

$$0 = A + \frac{4}{5}$$

Finally,

$$y(t) = \frac{4}{5}\left[e^{3t} - e^{-2t}\right] u(t)$$

Different auxiliary conditions lead to different solutions

More understanding about causality

1. The initial rest can guarantee the linear system since zero-input of linear system must lead to zero-output (non-zero auxiliary conditions may lead to incrementally linear system)
2. The initial rest can also guarantee the time-invariance and causality of the system as can be intuitively explained, (can be proved by HW 2.33, 2.34)

- Some more: the homogenous solution to

$$\sum_{k=0}^{N} a_k \frac{d^k y_h(t)}{dt^k} = 0$$

has the form $y_h(t) = \sum_{k=1}^{N} C_k e^{\lambda_k t}$, where

– λ_k is the solution to $\sum_{k=0}^{N} a_k \lambda^k = 0$

Assume roots are all different and real

– C_k is unknown and should be determined from auxiliary conditions, e.g., initial rest.

Check Prof. Zheng's book (or math books) for conjugate roots, duplicates roots, as well as their combinations

LTI System Described by Difference Equations

- System be described by the linear constant-coefficient difference equations (LCCDE)

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

  - Particular solution plus homogeneous solution

$$y[n] = y_p[n] + y_h[n]$$

Natural response

where

$$\sum_{k=0}^{N} a_k y_h[n-k] = 0$$

  - Also need auxiliary conditions to make $y[n]$ explicit.

- Alternative solution:

$$y[n] = \frac{1}{a_0}\left\{\sum_{k=0}^{M} b_k x[n-k] - \sum_{k=1}^{N} a_k y[n-k]\right\}$$

Recursive equation

Clearly, auxiliary condition $y[n-1], \ldots, y[n-N]$ is needed

– Starting from $y[0]$, we know $y[-1], \ldots, y[-N]$ are needed

– For $N = 0$

$$y[n] = \sum_{k=0}^{M} \left(\frac{b_k}{a_0}\right) x[n-k]$$

Non-recursive equation

$$h[n] = \begin{cases} \frac{b_n}{a_0}, & 0 \le n \le M \\ 0, & \text{otherwise} \end{cases}$$

Finite impulse response (FIR) system

- When $N \geq 1$, check example

$$y[n] - \frac{1}{2}y[n-1] = x[n] \quad x[n] = \delta[n]$$

With initial rest

  - Since $x[n] = 0, \ n \leq -1$, the initial rest says

$$y[n] = 0, \ n \leq -1$$

    and we take $y[-1] = 0$. So

- When $N \geq 1$, check example

$$y[n] - \frac{1}{2}y[n-1] = x[n] \quad x[n] = \delta[n]$$

With initial rest

  - Since $x[n] = 0, \ n \leq -1$, the initial rest says

$$y[n] = 0, \ n \leq -1$$

and we take $y[-1] = 0$. So

$$y[0] = x[0] + \frac{1}{2}y[-1] = 1$$

$$y[1] = x[1] + \frac{1}{2}y[0] = \frac{1}{2}$$

$$y[2] = x[2] + \frac{1}{2}y[1] = \frac{1}{4}$$

$$\vdots$$

$$y[n] = x[n] + \frac{1}{2}y[n-1] = \frac{1}{2^n}$$

Infinite impulse response (IIR) system

$$h[n] = \frac{1}{2^n}u[n]$$

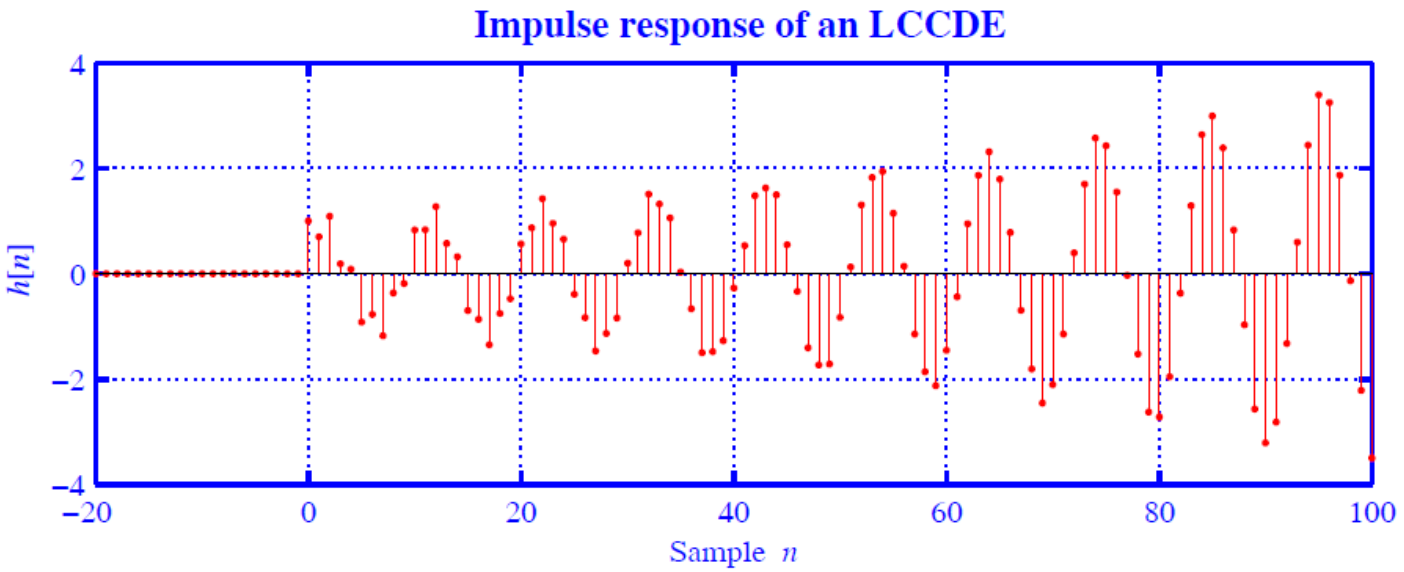
Given an LCCDE

$$y[n] - 0.7y[n-1] - 0.6y[n-2] + y[n-3] = x[n], \quad \forall n.$$

Find the impulse response $h[n]$ of the system.

✂ Solution: (in MATLAB© syntax):

```
a=[1,-0.7,-0.6,1];
b=[1];
n=[-20:100];
x=[(n)==0];
h=filter(b,a,x);
```

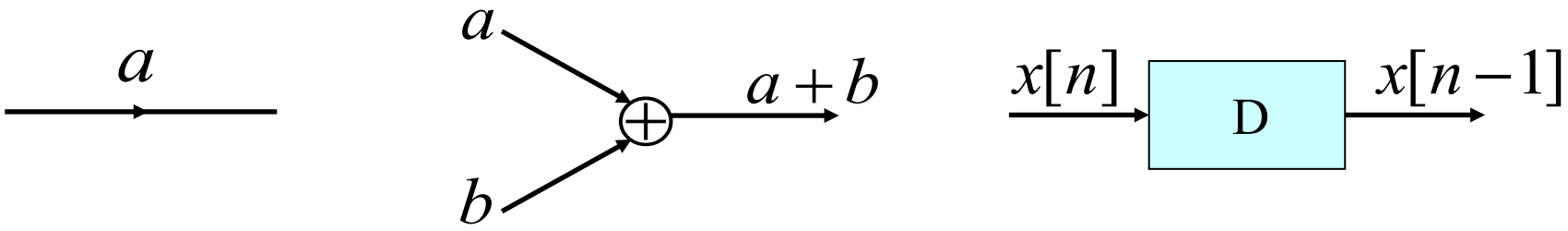


Block Diagram Representation:

- LCCDE is composed by elementary operations.
  - Possible to be represented by block diagram
  - Better understanding
  - Easy for simulation or implementation.
- Discrete time LCCDE

$$y[n] = \frac{1}{a_0}\left\{\sum_{k=0}^{M} b_k x[n-k] - \sum_{k=1}^{N} a_k y[n-k]\right\}$$

  - Tree basic operations: multiplication, addition, and delay

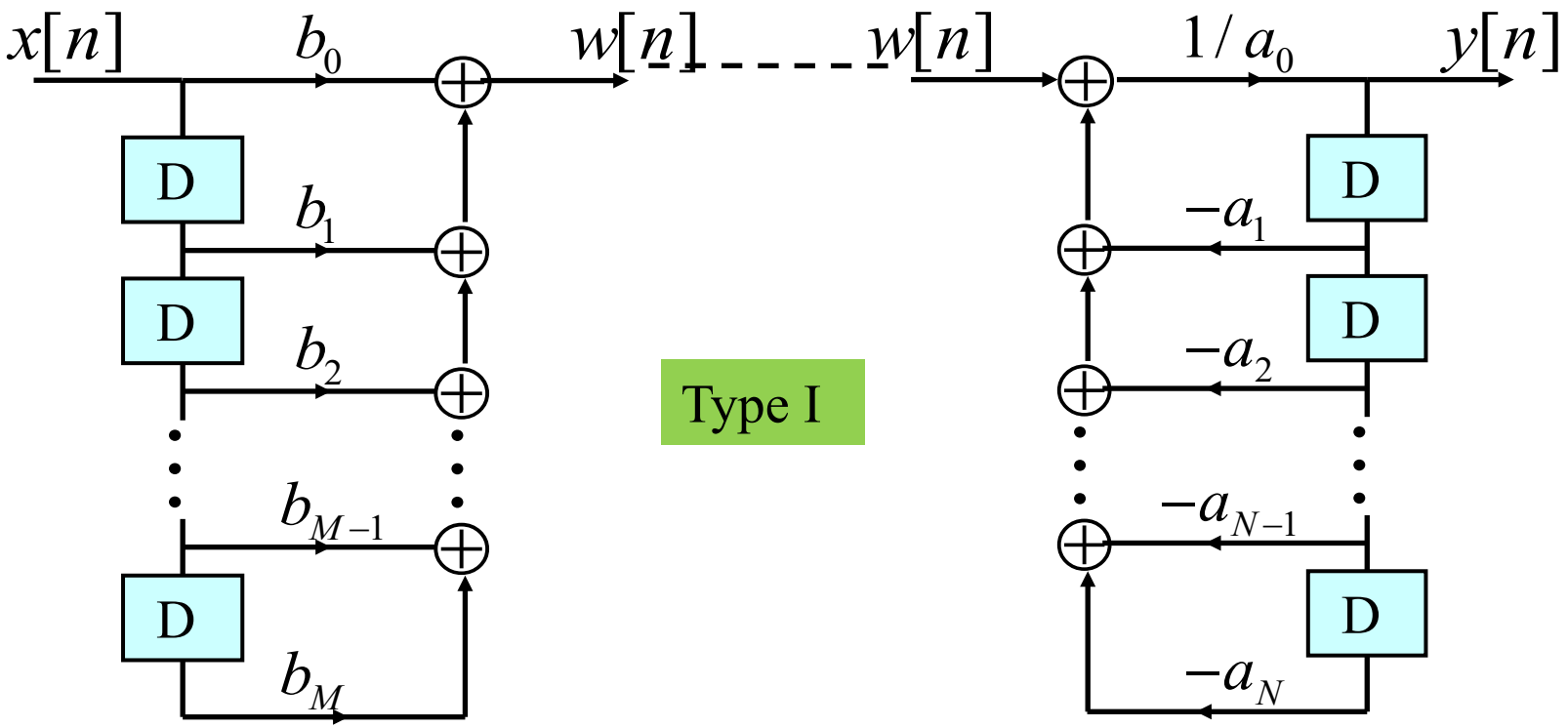


$$y[n] = \frac{1}{a_0}\left\{\sum_{k=0}^{M} b_k x[n-k] - \sum_{k=1}^{N} a_k y[n-k]\right\}$$

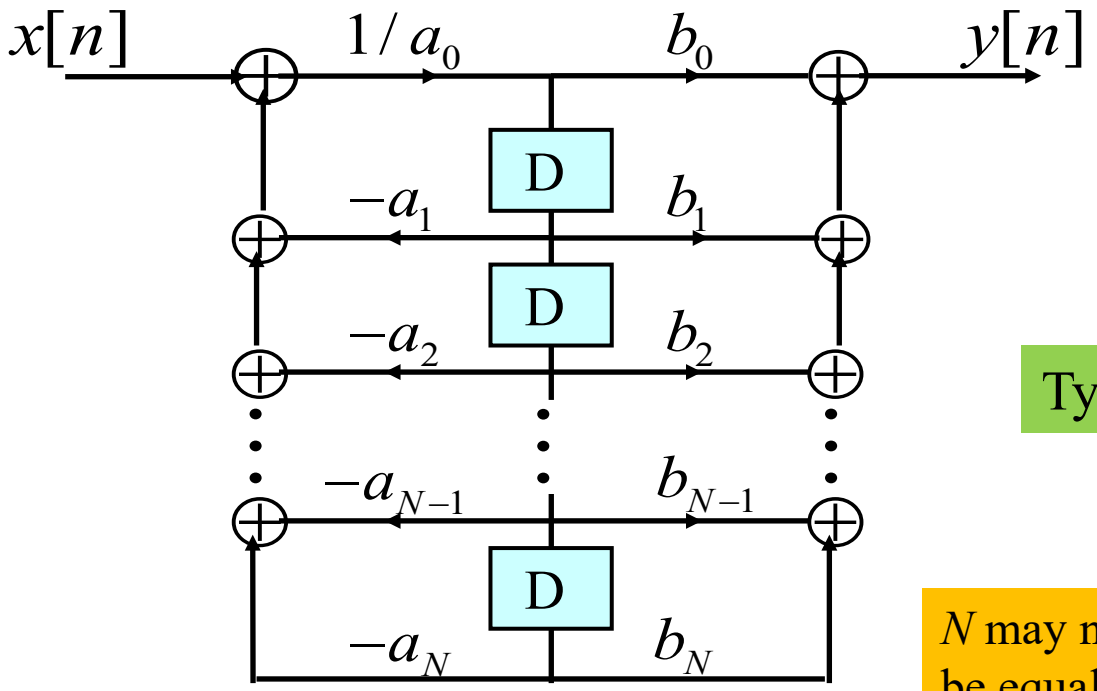
- Let

$$w[n] = \sum_{k=0}^{M} b_k x[n-k] \qquad y[n] = \frac{1}{a_0}\left\{w[n] - \sum_{k=1}^{N} a_k y[n-k]\right\}$$

  - Then



- Can be viewed as the cascade of two sub-systems. Hence, one can switch their order according to the associate property, and then get



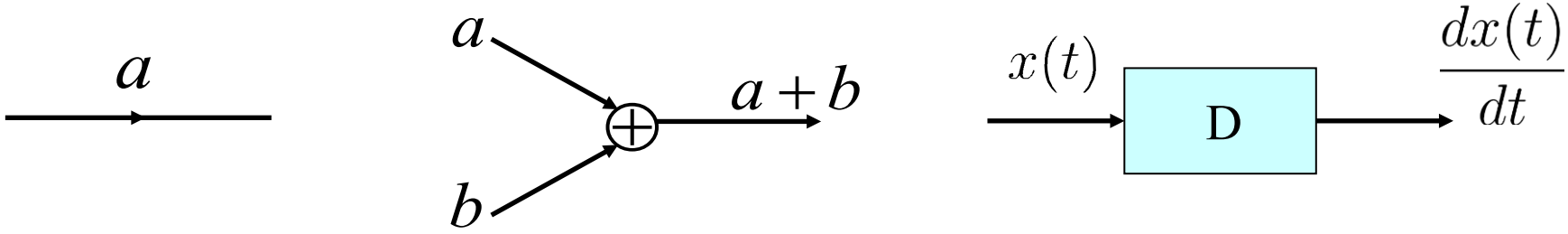
Type II

N may not necessarily be equal to *M*

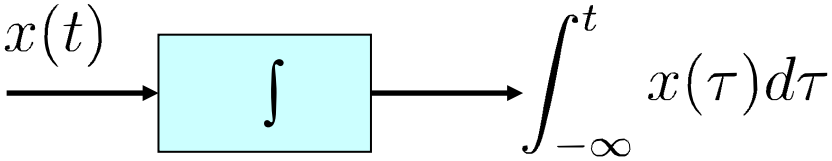
- Continuous-time LCCDE

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}$$

  - Tree basic operations: multiplication, addition, and differentiation



  - Since differentiator is difficult to implement and is sensitive to errors and noise, it is replaced by integrator



$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}$$

- Integrate N times on both sides, we obtain

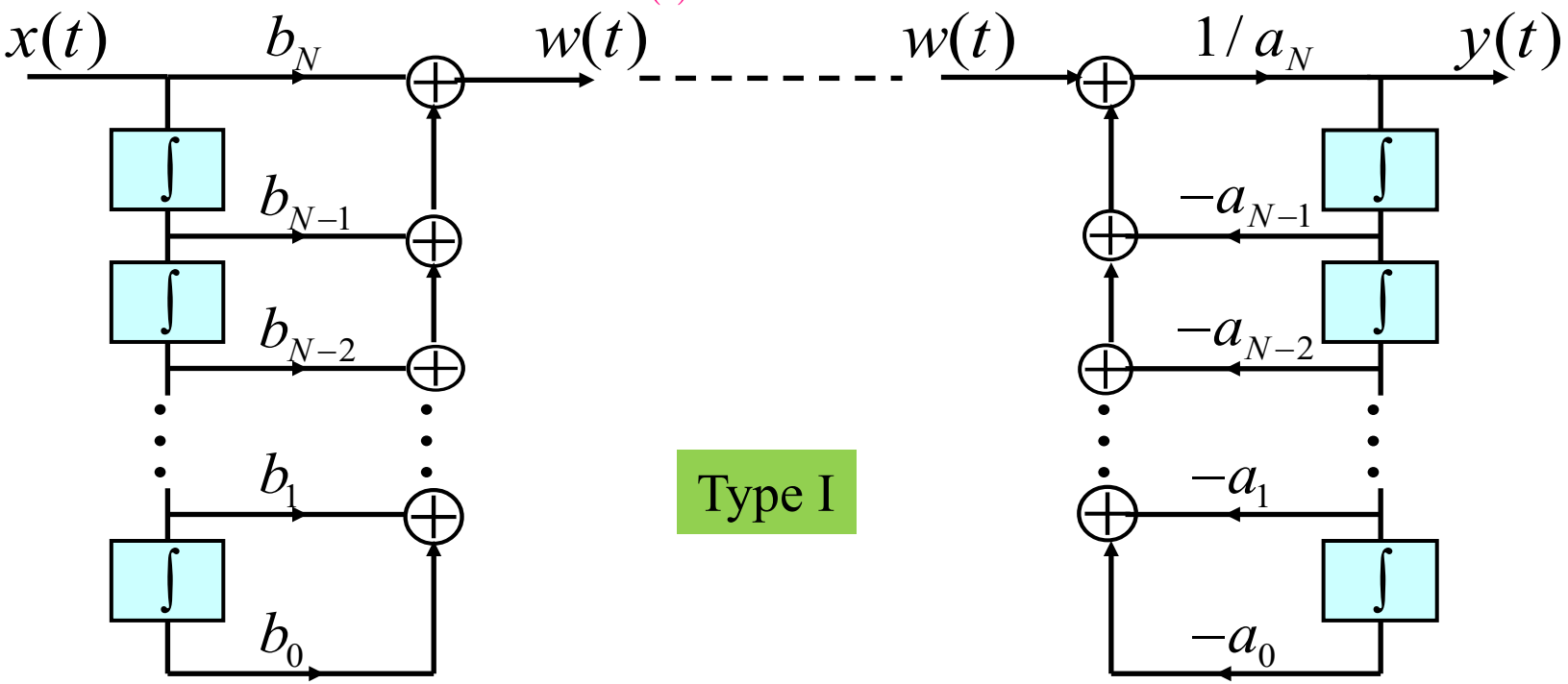
$$\sum_{k=0}^{N} a_k y_{(N-k)}(t) = \sum_{k=0}^{M} b_k x_{(N-k)}(t)$$

where $y_{(i)}(t)$ represents the i-fold integration

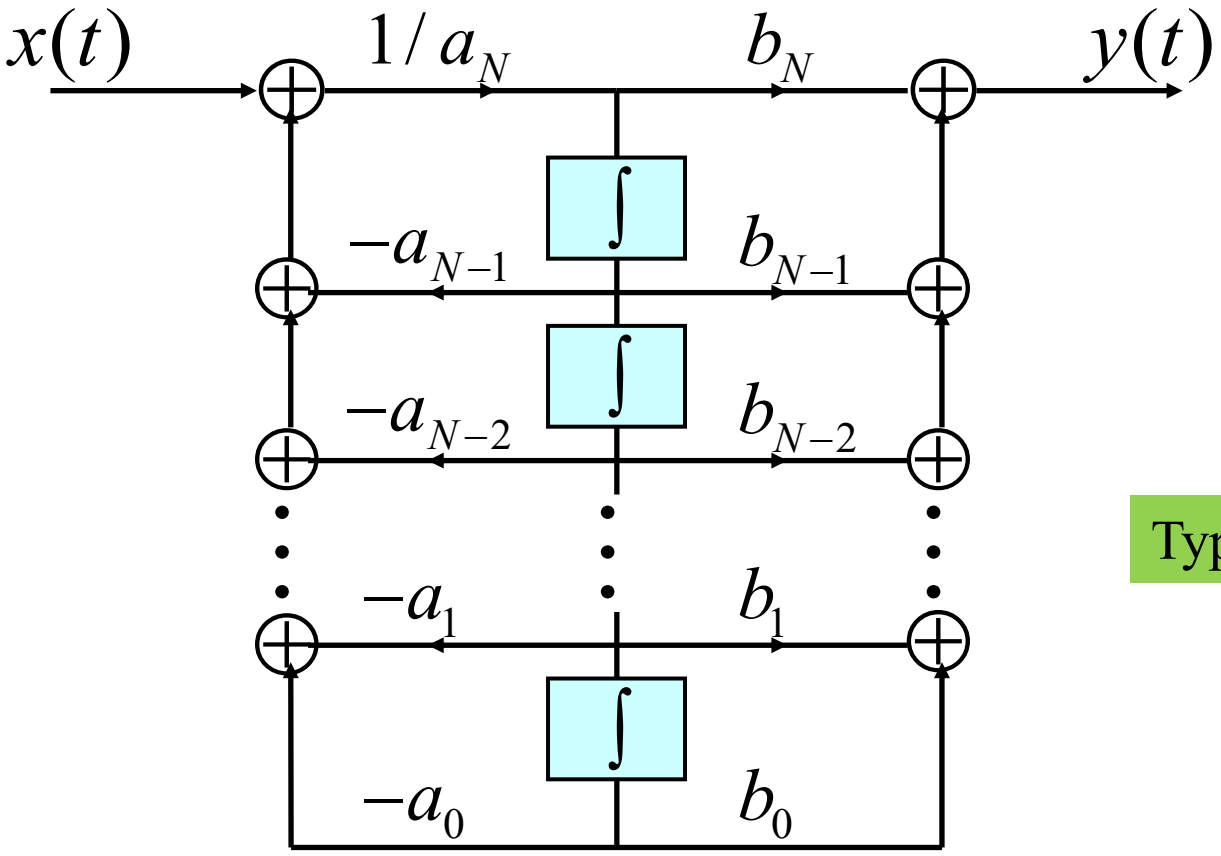
– Similarly as discrete time case

$$\Longrightarrow \quad y(t) = \frac{1}{a_N} \left(\sum_{k=0}^{M} b_k x_{(N-k)}(t) - \sum_{k=0}^{N-1} a_k y_{(N-k)}(t) \right)$$

$$y(t) = \frac{1}{a_N}\left[\underbrace{\sum_{k=0}^{N} b_k x_{(N-k)}(t)}_{w(t)} - \sum_{k=0}^{N-1} a_k y_{(N-k)}(t)\right]$$



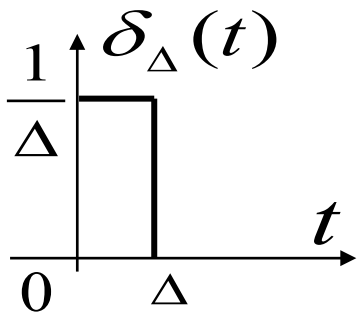
- According to associate property, let's switch

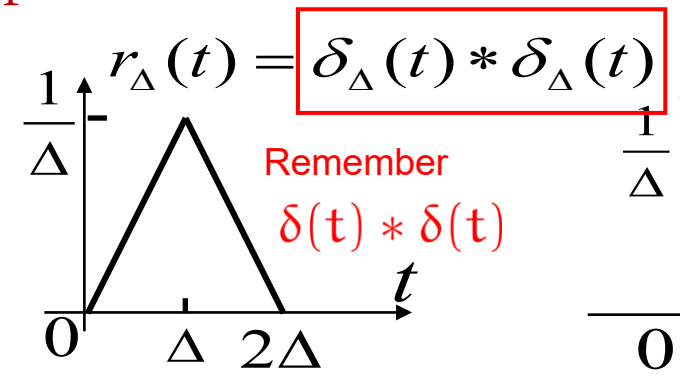


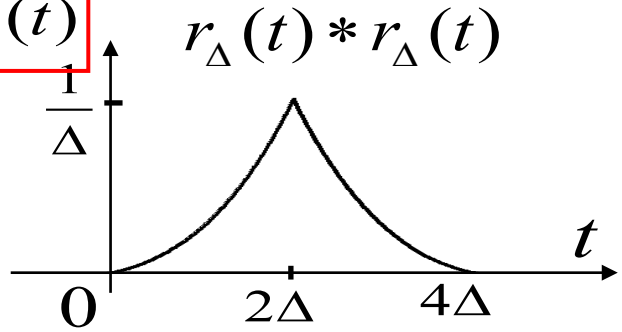
Type-II

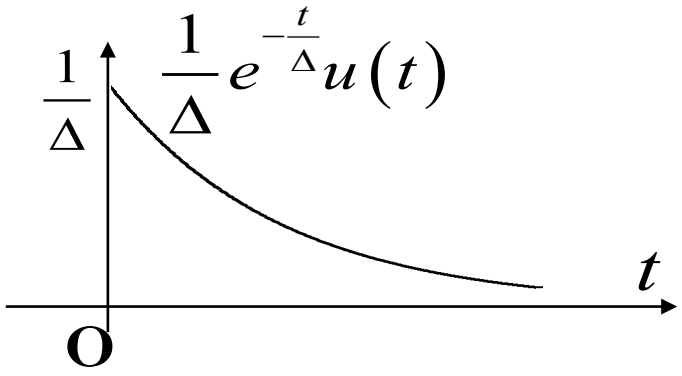
Singularity Functions

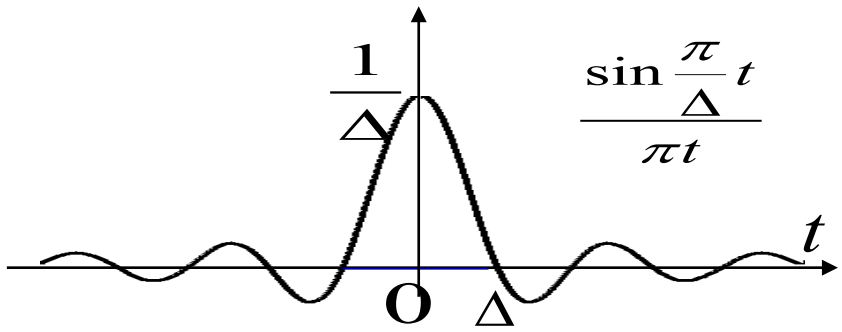
- Previously when we define $\delta(t)$, we consider an square impulse $\delta_\Delta(t)$ with unit area and take $\Delta \to 0$.
  - However, such definition is not clear, as many functions behave like an impulse











- Hence, $\delta(t)$ is special function and is called as singularity function, which is often defined by integration or convolution.
- Definition by integration: $x(0) = \int_{-\infty}^{\infty} x(t)\delta(t)dt$
  - If $x(t) = 1$, then $\int_{-\infty}^{\infty} \delta(t)dt = 1$. Similarly $\int_{-\infty}^{\infty} \delta(t - t_0)dt = 1$
  - From $\int_{-\infty}^{\infty} x(t)\delta(t)dt = x(0) = \int_{-\infty}^{\infty} x(0)\delta(t)dt$

not strict

$\Rightarrow x(t)\delta(t) = x(0)\delta(t) \Rightarrow t\delta(t) = 0$

  - Similarly $x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0)$

– If $tx_1(t) = tx_2(t)$, then $x_1(t) = x_2(t) + k\delta(t)$

– Prove even property: For any $x(t)$, there is

$$x(0) = \int_{-\infty}^{\infty} x(t)\delta(t)dt = \int_{-\infty}^{\infty} x(-t)\delta(t)dt = \int_{-\infty}^{\infty} x(t)\delta(-t)dt$$

$$x(t)\delta(t) = x(t)\delta(-t) \Rightarrow \delta(t) = \delta(-t)$$

– Moreover,

Strictly, we should understanding the equality from the equivalent integrating, but not from the individual function!!

$$\int_{-\infty}^{\infty} x(t)\delta(at)dt = \frac{1}{a}\int_{-\infty}^{\infty} x(\frac{\tau}{a})\delta(\tau)d\tau = \frac{1}{a}x(0) = \frac{1}{a}\int_{-\infty}^{\infty} x(t)\delta(t)dt \quad (a > 0)$$

$$\int_{-\infty}^{\infty} x(t)\delta(at)dt = -\frac{1}{a}\int_{-\infty}^{\infty} x(t)\delta(t)dt \quad (a < 0)$$

$$\Rightarrow \delta(at) = \frac{1}{|a|}\delta(t)$$

- Definition by convolution

$$x(t) = x(t) * \delta(t) = \int_{-\infty}^{+\infty} \delta(\tau) x(t-\tau) d\tau$$

  - If $x(t) = 1$, then $\int_{-\infty}^{+\infty} \delta(\tau) d\tau = 1$

  - In fact, define $g(\tau) = x(t-\tau)$, then

$$\begin{cases} x(t) = \int_{-\infty}^{+\infty} \delta(\tau) x(t-\tau) d\tau = \int_{-\infty}^{+\infty} \delta(\tau) g(\tau) d\tau \\ x(t) = g(\tau = 0) = g(0) \end{cases}$$

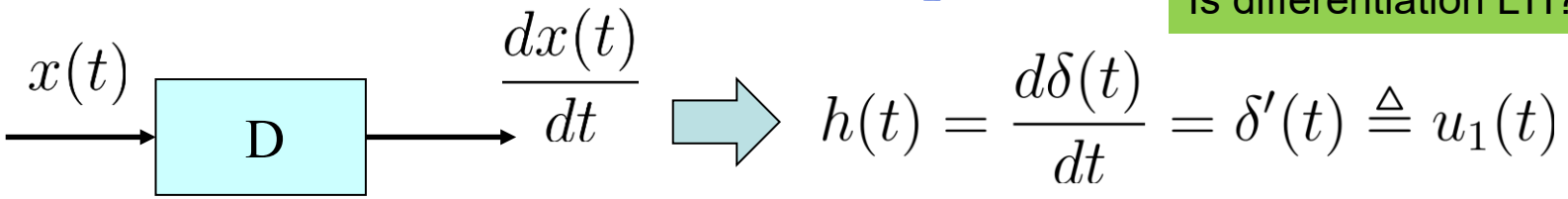
$$\Rightarrow \quad g(0) = \int_{-\infty}^{+\infty} \delta(\tau) g(\tau) d\tau$$

Definition by integration

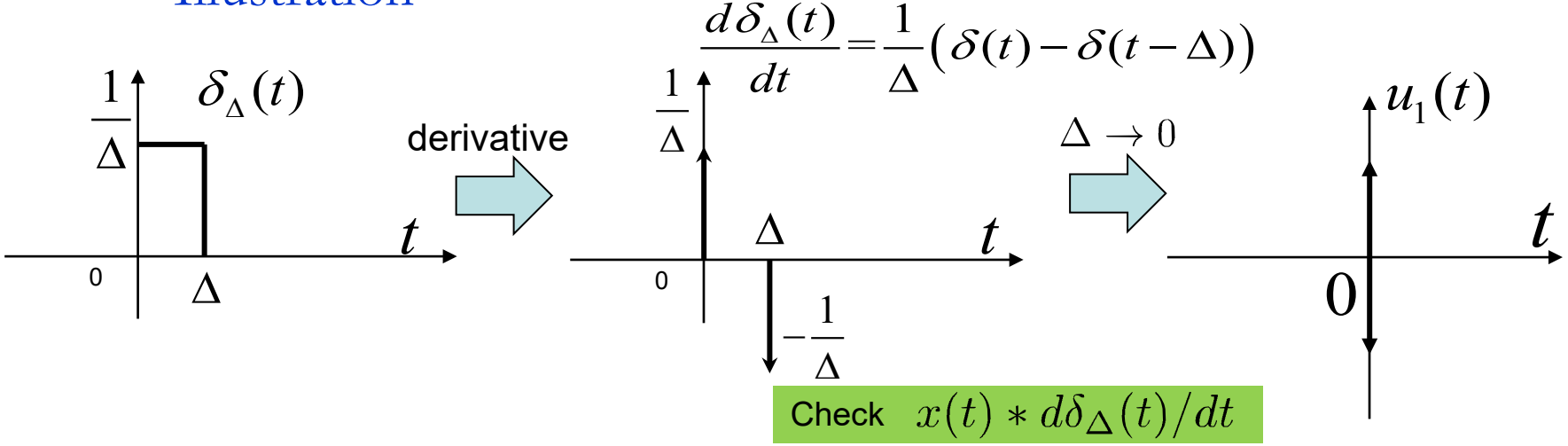
Unit Doublets and Other Singularity Functions

- Consider system with derivative output

Is differentiation LTI?



  - Illustration



  - Double impulses at $t = 0$ which are mirror imaged with their amplitude of infinite.

– Let $x(t) = 1$, we know

$$\frac{dx(t)}{dt} = \int_{-\infty}^{+\infty} u_1(\tau) d\tau = 0$$

How to prove $u_1(t)$ is odd function?

– Let $x(t) = 1$, we know

$$\frac{dx(t)}{dt} = \int_{-\infty}^{+\infty} u_1(\tau)d\tau = 0$$

– Let $x(-t)$ pass the system

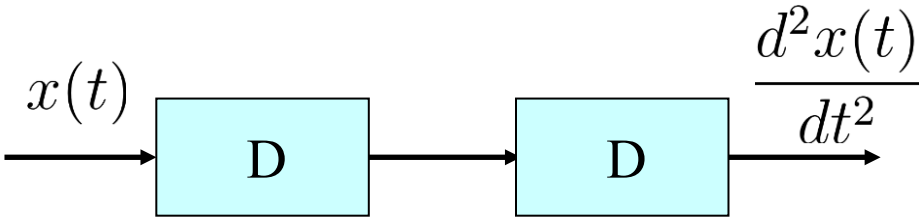
$$\int_{-\infty}^{+\infty} x(\tau - t)u_1(\tau)d\tau = \frac{dx(-t)}{dt} = -x'(-t)$$

$$\Rightarrow \quad \int_{-\infty}^{+\infty} x(\tau)u_1(\tau)d\tau = -x'(0)$$

– For any $x(t)$, there is

$$\left.\begin{aligned} &\int_{-\infty}^{+\infty} x(\tau)u_1(\tau)d\tau = -x'(0) \\ &\text{Similarly, input } x(t), \text{ we obtain} \\ &\int_{-\infty}^{+\infty} x(-\tau)u_1(\tau)d\tau = x'(0) \end{aligned}\right\} \begin{aligned} &\int_{-\infty}^{+\infty} x(\tau)u_1(\tau)d\tau = -\int_{-\infty}^{+\infty} x(-\tau)u_1(\tau)d\tau \\ &= \int_{-\infty}^{+\infty} x(\tau)(-u_1(-\tau))d\tau \end{aligned}$$

- Similarly



$$\Rightarrow \quad h(t) = \frac{d^2\delta(t)}{dt^2} = \delta''(t) \triangleq u_2(t) = u_1(t) * u_1(t)$$

- Similarly

$$u_k(t) = \frac{d^k\delta(t)}{dt^k} = \underbrace{u_1(t) * \ldots * u_1(t)}_{k \text{ times}}$$

$\delta(t) = u_0(t)$

- Similarly

$$u_{-k}(t) = \underbrace{\int_{-\infty}^{t} \ldots \int_{-\infty}^{t}}_{k \text{ times}} \delta(\tau)d\tau = \overbrace{u(t) * \ldots * u(t)} \quad u(t) = u_{-1}(t)$$

See closed form in textbook!

$u_k(t) * u_r(t) = u_{k+r}(t)$

Summary

- Analysis of LTI System LTI——convolutional sum and convolutional integral
- Description of LTI system
  - With impulse response
  - With LCCDE and initial rest
  - With block diagram
- Property of LTI system as well as its impulse response
  - Memory, causality, stability, invertibility
  - When system is cascade or parallel connected
- Singularity functions

Homework #2

- 2.3, 2.8, 2.16, 2.22(d)(e), 2.24, 2.31, 2.38, 2.39, 2.40, 2.48, 2.51
- Try whenever you think you can handle.
- Try finish all questions after the chapter.
- Homework is always due 1 week after each Chapter is finished.